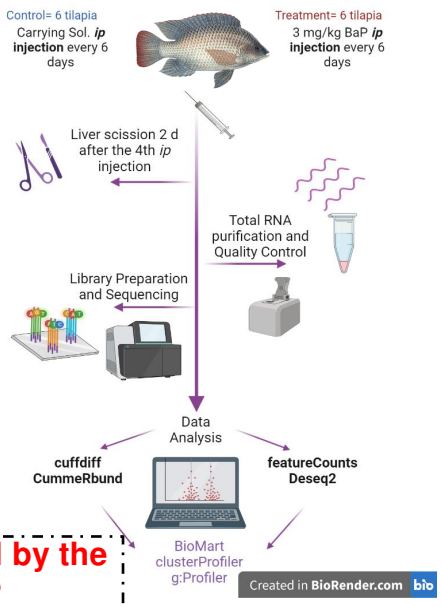
Comparison of two methods for detecting differential expression in a RNA-seq Tilapia study

Reyna Cristina Colli-Dula, Nacira Anahi Albornoz-Abud

Benzo(a)pyrene (BaP) is a toxic hydrocarbon present in **aquatic systems** derived from various anthropogenic activities and can be **bioaccumulated** in aquatic organisms.

Early responses detection to BaP exposure is important at the molecular level in aquatic organisms using RNA-sequencing techniques.

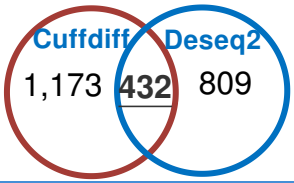
Although, many data analysis methods developed for RNA-Seq analysis its not clear **which criteria to choose to select the most appropriate method**, especially for a basic user without a solid statistical or computational background.



Could results be highly affected by the use of different packets?

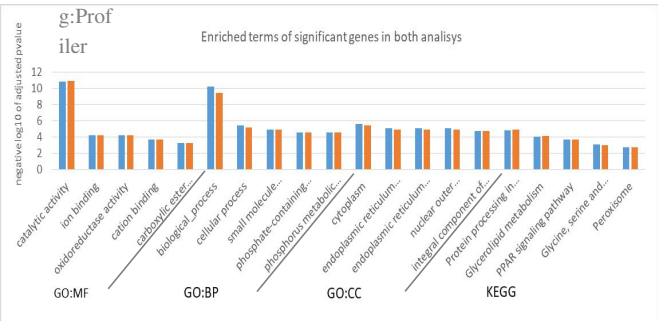
We performed a comparison of **two widely used Bioconductor packages (Cuffdiff and DESeq2)** to determine the differential expressed genes (DEG) and the enrichment analysis between sample groups **(BioMart, ClusterProfiler and g:Profiler)**

We identified **432** transcripts having the same patterns of expression between the two methods.

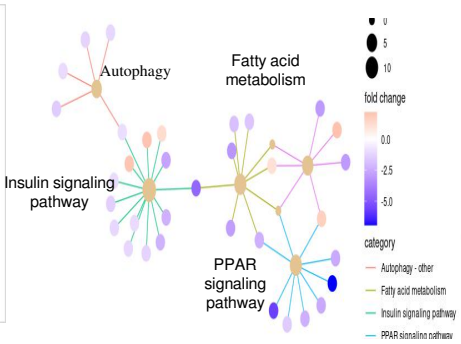


DEG in both analysis				
	Deseq2		Cuffdiff	
Gene	log2FC	padj	log2FC	padj
me1	-6.52	1E-43	-6.93	0.0017
morn3	-6.44	3E-23	-6.04	0.0152
fabp7a	-5.70	3E-11	-5.66	0.0017
bhlhe40	-4.66	3E-14	-4.89	0.0017
fasn	-4.31	6E-19	-4.63	0.0017
acaca	-4.17	6E-11	-4.56	0.0017
lrrc39	-3.81	6E-12	-4.54	0.0052
penk	3.12	7E-05	3.87	0.0041
tspan2b	3.16	4E-07	3.13	0.0017
AQP4	3.53	0.0004	3.52	0.0087
akr1a1a	4.73	1E-16	5.41	0.0017
vwa11	5.31	4E-07	5.37	0.0017

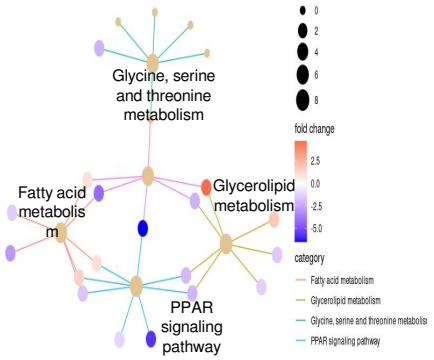
We found that the enriched **GO categories for biological processes were consistent using the ClusterProfiler package and g:Profiler plaftform website**. At the pathway level, **both packages show similar results.**

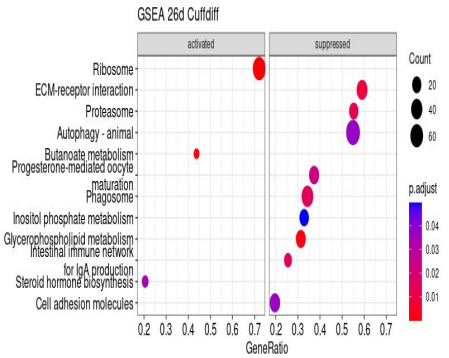


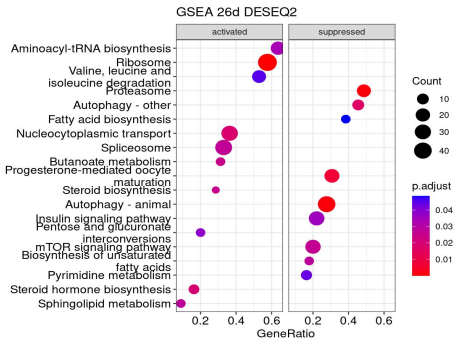
ORA Cuffdiff



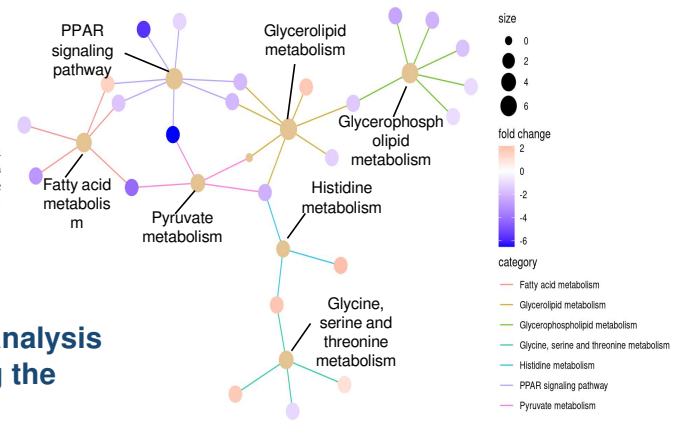
ORA Deseq2







ORA of significant genes in both analyzes



Conclussion: Our study shows and demonstrates how data analysis tools can markedly affect the outcome of results, highlighting the importance of taking into account the best option.

REFERENCES

- Colli-Dula, R.C., Fang, X., Moraga-Amador, D., Albornoz-Abud, N., Zamora-Bustillos, R., Conesa, A., Zapata-Perez, O., Moreno, D., Hernandez-Nunez, ˜ E., 2018. Transcriptome analysis reveals novel insights into the response of low-dose benzo(a) pyrene exposure in male tilapia. Aquat. Toxicol. 201, 162–173.
- Padros, J., Pelletier, E., Ribeiro, C.O., 2003. Metabolic interactions between low doses of benzo[a]pyrene and tributyltin in arctic charr (Salvelinus alpinus): a long-term in vivo study. Toxicol. Appl. Pharmacol. 192 (1), 45–55

@ColliDula

rcolli.dula@cinvestav.mx